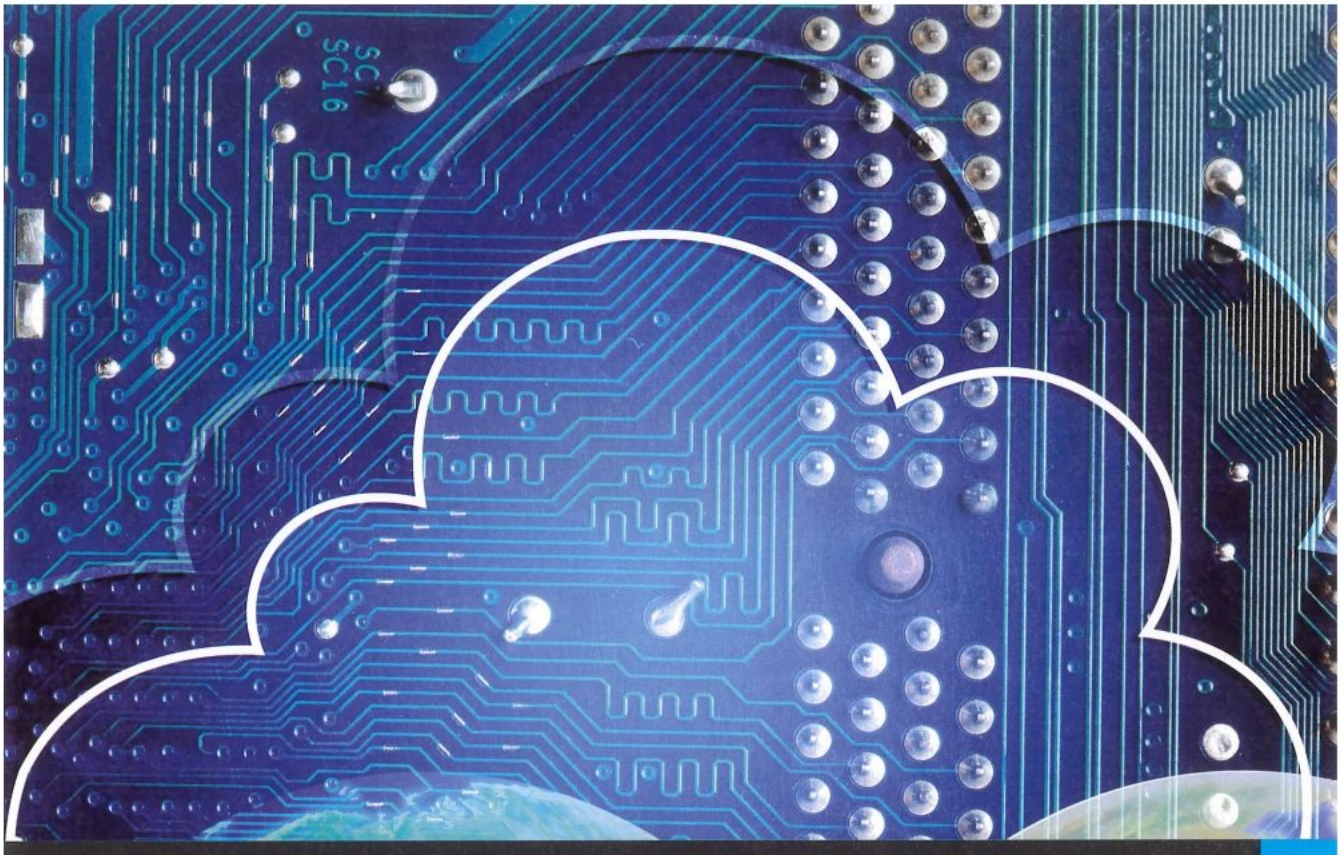
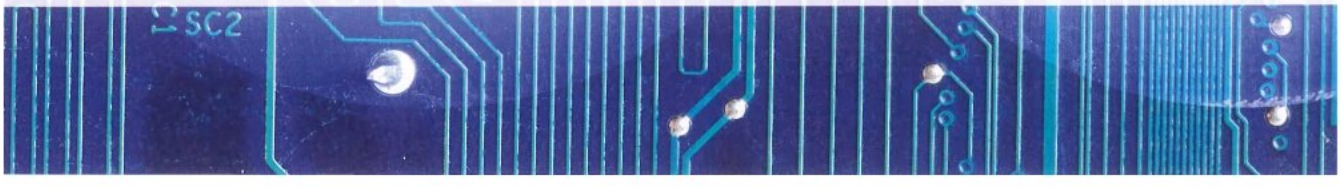


Chapter 9:
Security, privacy and data integrity

Learning objectives

By the end of this chapter you should be able to:

- explain the difference between the terms security, privacy and integrity of data
- show appreciation of the need for both the security of data and the security of the computer system
- describe security measures designed to protect computer systems, ranging from the stand-alone PC to a network of computers
- show understanding of the threats to computer and data security posed by networks and the Internet
- describe methods that can be used to restrict the risks posed by threats
- describe security measures designed to protect the security of data
- describe how data validation and data verification help protect the integrity of data
- describe and use methods of data validation
- describe and use methods of data verification during data entry and data transfer.



9.01 Definitions of data integrity, privacy and security

Data integrity

It is easy to define integrity of data but far less easy to ensure it. Only accurate and up-to-date data has **data integrity**. Any person or organisation that stores data needs it to have integrity. We will discuss ways to achieve data integrity in this chapter, and also in Chapter 11 (Sections 11.01 & 11.02).

KEY TERMS

Data integrity: a requirement for data to be accurate and up to date

Data privacy: a requirement for data to be available only to authorised users

Data privacy

Data privacy is about keeping data private rather than allowing it to be available in the public domain. The term may be applied to a person or an organisation. Every individual has an almost limitless amount of data associated with their existence. Assuming that an individual is not engaged in criminal or subversive activities, he or she should be in control of which data about himself or herself is made public and which data remains private. An organisation can have data that is private to the organisation, such as the minutes of management meetings, but we will not discuss this here.

For an individual there is little chance of data privacy if there is not a legal framework in place to penalise offenders who breach this privacy. This framework is provided by a **data protection law**. The following aspects should be noted about such laws.

- The major focus relates to personal, therefore private, data that an individual supplies to an organisation.
- The data is supplied to allow the organisation to use it but only for purposes understood and agreed by the individual.
- Data protection laws oblige organisations to ensure the privacy and the integrity of this data.
- Unfortunately, having laws does not guarantee adherence to them but they do act as a deterrent if wrong-doers can be subject to legal proceedings.

KEY TERM

Data protection law: a law that relates to data privacy

Discussion Point:

What data protection laws are in place in your country? Are you familiar with any details of these laws?

Data protection normally applies to data stored in computer systems with the consent of the individual. Should these laws be extended to cover storage of data obtained from telephone calls or search engine usage?

Data security

Data are secure if they are available for use when needed and the data made available are the data that were stored originally. The security of data has been breached if any data have been lost or corrupted.

Data security must be achieved before either data integrity or data privacy can be achieved, but data security does not by itself guarantee either data integrity or data privacy. One of the requirements for protection of data is the security of the system used to store the data. System security does not just protect data. There are two primary aims of system security measures:

- to ensure that the system continues to carry out the tasks users need
- to ensure that only authorised users have access to the system.

KEY TERM

Data security: a requirement for data to be available for use when needed, ensures that only authorised users have access to the system and data can be recovered if lost or corrupted.

9.02 Threats to the security of a computer system and of the data stored in it

The threats to the security of a system include the following types:

- individual user not taking appropriate care
- internal mismanagement
- natural disasters
- unauthorised intrusion into the system by an individual
- malicious software entering the system.

TIP

Try to keep a perspective with relation to hacking; it is only one of many security issues.

Threats to computer and data security posed by networks and the Internet

We are all continuously at risk from security threats to systems we use ourselves and to all of the systems used by organisations upon which we rely. The dominant factor is that none of these systems are stand-alone; all are connected to networks and through these networks to the Internet. One cause of concern is the hacker who is someone intent on gaining unauthorised access to a computer system. A hacker who achieves this aim might gain access to private data. Alternatively, a hacker might cause problems by deleting files or causing problems with the running of the system. The other major cause of concern is malicious software entering the system.

KEY TERM

Malware: malicious software that has the intention of causing harm to a system or its contents

Types of malware

Malware is the everyday name for malicious software. It is software that is introduced into a system for a harmful purpose. One category of malware is where program code is introduced to a system. The various types of malware-containing program code are:

- virus: tries to replicate itself inside other executable code
- worm: runs independently and transfers itself to other network hosts
- logic bomb: stays inactive until some condition is met

- Trojan horse: replaces all or part of a previously useful program
- spyware: collects information and transmits it to another system
- bot: takes control of another computer and uses it to launch attacks.

The differences between the different types are not large and some examples come into more than one of these categories. The virus category is often subdivided according to the software that the virus attaches itself to. Examples are boot sector viruses and macro viruses.

Malware can also be classified in terms of the activity involved:

- phishing: sending an email or electronic message from an apparently legitimate source requesting confidential information
- pharming: setting up a bogus website which appears to be a legitimate site
- keylogger: recording keyboard usage by the legitimate user of the system.

Question 9.01

Carry out some research to find some examples of how phishing and pharming might be attempted.

System vulnerability arising from user activity

Many system vulnerabilities are associated directly with the activities of legitimate users of a system. Two examples which do not involve malware are as follows.

- The use of weak passwords and particularly those which have a direct connection to the user. A poor choice of password gives the would-be hacker a strong chance of guessing the password and thus being able to gain unauthorised access.
- A legitimate user not recognising a phishing or pharming attack and, as a result, giving away sensitive information.

A legitimate user with a grievance might introduce malware deliberately. More often, malware is introduced accidentally by the user. Typical examples of actions that might introduce malware are:

- attaching a portable storage device
- opening an email attachment
- accessing a website
- downloading a file from the Internet.

Vulnerability arising from within the system itself

Systems themselves often have security weaknesses. The following are examples of this.

1. Operating systems often lack good security. Over time, there is a tendency for operating systems to increase in complexity, which leads to more opportunities for weak security. Operating systems have regular updates, often because of a newly discovered security vulnerability.
2. In the past, commonly used application packages allowed macro viruses to spread, but this particular problem is now largely under control.
3. A very specific vulnerability is buffer overflow. Programs written in the C programming language, of which there are very many, do not automatically carry out array bound checks. A program can be written to deliberately write code to the part of memory that is outside the address range defined for the array, set up as a buffer. The program overwrites what is stored there so when a later program reads this overwritten section it will not execute as it should. Sometimes this only causes minor disruption, but a cleverly designed program can permit an attacker to gain unauthorised access to the system and cause serious problems.